

MORUYA AP, Australia

WMO: 959370

Lat: 35.9003S

Lon: 150.1436E

Elev: 5

StdP: 101.27

Time Zone: 10.00 (AUS)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	1.2	2.3	-3.1	2.9	12.6	-1.6	3.3	11.6	9.1	14.3	8.1	14.8	1.3	250	0.450

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.1	29.1	19.0	26.5	20.1	25.2	20.2	22.7	25.3	22.1	24.6	21.5	24.0	5.5	330

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.9	16.5	23.8	21.2	15.9	23.4	20.6	15.3	23.0	67.2	25.6	64.7	24.7	62.6	24.0	27.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.6	7.6	DB	-1.6	37.7	0.9	3.6	-2.3	40.2	-2.8	42.3	-3.3	44.3	-4.0	46.9	(4)
(5)			WB	-1.8	23.8	0.9	1.1	-2.4	24.6	-2.9	25.2	-3.4	25.8	-4.1	26.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.4	20.3	20.2	18.9	16.5	13.4	11.5	10.4	11.0	12.9	15.0	16.8	18.6
	DBStd	4.16	2.33	2.09	2.06	2.06	2.12	2.06	2.15	2.38	2.72	2.96	2.61	2.25
	HDD10.0	47	0	0	0	0	1	8	20	14	3	1	0	0
	HDD18.3	1283	6	4	17	62	153	206	246	228	164	113	59	24
	CDD10.0	2026	318	286	276	194	107	52	32	44	92	154	205	266
	CDD18.3	220	66	56	35	6	1	0	0	0	2	9	14	31
	CDH23.3	1009	276	207	119	41	9	0	1	3	47	111	91	106
CDH26.7	241	63	35	16	11	1	0	0	0	9	44	35	27	
Wind	WSAvg	3.3	3.8	3.4	3.4	3.0	2.8	2.9	2.8	3.1	3.2	3.6	3.7	3.7
Precipitation	PrecAvg	1014	92	100	117	98	80	88	50	63	66	90	92	81
	PrecMax	1755	362	576	571	506	260	514	224	310	197	525	302	247
	PrecMin	468	10	2	6	2	1	0	0	1	3	7	9	10
	PrecStd	320	70	121	108	110	65	101	53	73	49	95	67	61
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	33.6	31.6	28.4	27.6	24.4	20.7	21.6	23.5	27.9	32.4	31.3	30.1
		MCWB	21.4	21.7	19.8	16.6	15.6	13.4	12.3	13.2	14.5	17.0	18.5	19.5
	2%	DB	27.3	26.5	25.6	23.9	21.9	18.8	18.9	20.6	24.1	26.0	25.2	25.6
		MCWB	21.5	22.1	20.9	18.0	15.5	13.1	11.7	12.4	14.0	15.5	18.6	20.4
	5%	DB	25.5	25.3	24.5	22.6	20.3	17.6	17.5	18.5	21.2	22.4	22.9	24.0
		MCWB	21.3	21.5	20.3	18.0	14.8	12.3	11.3	11.8	13.6	15.8	18.8	20.0
	10%	DB	24.5	24.4	23.5	21.5	18.9	16.6	16.2	17.1	18.9	20.4	21.6	23.0
		MCWB	20.8	20.8	19.6	17.3	13.9	11.9	10.8	11.3	13.8	16.1	18.1	19.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.4	23.7	23.1	20.7	18.2	16.0	14.9	15.5	17.0	19.3	21.3	22.2
		MCDB	27.1	26.1	24.5	23.3	21.0	18.0	17.4	19.3	21.2	22.6	24.4	25.3
	2%	WB	22.6	22.8	21.8	19.6	16.8	14.6	13.5	14.2	16.1	18.3	20.1	21.4
		MCDB	25.5	25.2	24.1	22.3	20.0	16.9	16.7	17.9	19.6	21.5	23.2	24.0
	5%	WB	22.0	22.1	21.1	18.8	15.8	13.7	12.5	13.3	15.3	17.5	19.3	20.7
		MCDB	24.9	24.5	23.5	21.5	18.9	16.0	15.7	16.7	18.9	20.9	22.2	23.2
	10%	WB	21.4	21.5	20.4	17.8	14.9	12.8	11.5	12.4	14.4	16.6	18.5	20.0
		MCDB	24.0	23.8	22.9	20.6	18.0	15.2	14.9	15.9	18.3	20.1	21.3	22.5
Mean Daily Temperature Range	5% DB	MDBR	8.1	7.7	8.4	9.9	11.5	10.4	12.1	12.3	11.9	10.6	9.1	8.1
		MCDBR	10.4	9.5	10.3	12.0	13.8	12.1	13.6	14.3	16.3	15.5	12.3	10.3
		MCWBR	5.5	5.4	6.2	7.2	8.2	7.3	7.9	8.3	7.7	7.1	6.7	5.8
	5% WB	MCDBR	8.8	8.3	8.7	10.3	11.6	9.8	11.9	12.1	12.6	11.5	10.5	8.6
		MCWBR	5.3	5.2	5.9	6.9	7.7	6.8	8.1	8.2	7.8	6.9	6.4	5.4
Clear-Sky Solar Irradiance	taub	0.423	0.383	0.355	0.329	0.312	0.299	0.290	0.299	0.333	0.351	0.374	0.382	
	taud	2.346	2.503	2.559	2.608	2.600	2.632	2.653	2.620	2.519	2.501	2.473	2.468	
	Ebn at Noon	914	929	919	888	847	835	867	908	926	951	954	958	
	Edh at Noon	132	109	96	80	71	64	66	78	97	108	116	118	
All-Sky Solar Radiation	RadAvg	6.50	5.60	4.79	3.72	2.89	2.23	2.63	3.49	4.75	5.71	6.14	6.67	
	RadStd	0.53	0.46	0.35	0.30	0.14	0.17	0.21	0.18	0.32	0.39	0.47	0.44	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (26 neighbors)		+0.39	N/A	N/A	N/A	N/A	N/A	N/A	-75	+125	+52

Nomenclature: See separate page